13.2.2 Recovering a failed virtual machine to another availability zone with the help of autoscaling

In the first part of the chapter, you used a CloudWatch alarm to trigger the recovery of a virtual machine that was running a Jenkins CI server in case of a failure. This mechanism launches an identical copy of the original virtual machine, if necessary. This is possible only in the same availability zone, because the private IP address and the EBS volume of a virtual machine are bound to a single subnet and a single availability zone. But suppose your team isn't happy about the fact that they won't be able to use the Jenkins server to test, build, and deploy new software in the case of a unlikely availability zone outage. You need a tool that will let you recover in another availability zone.

Failing over into another availability zone is possible with the help of *autoscaling*. You can find the CloudFormation template for this example on GitHub and on S3. Download a snapshot of the repository at https://github.com/AWSinAction/code3/archive/main.zip. The file we're talking about is located at chapter13/multiaz.yaml. On S3, the same file is located at https://s3.amazonaws.com/awsinaction-code3/chapter13/multiaz.yaml.

Autoscaling is part of the EC2 service and helps you to ensure that a specified number of EC2 instances is running, even when availability zones become unavailable. You can use autoscaling to launch a virtual machine and make sure a new instance is started if the original instance fails. You can use it to start virtual machines in multiple subnets. For example, if you have an outage of an entire availability zone, a new instance can be launched in another subnet in another availability zone.

Execute the following command to create a virtual machine that can recover in another availability zone if necessary. Replace `$Password` with a password consisting of 8–40 characters:

```
$ aws cloudformation create-stack --stack-name jenkins-multiaz \
➥ --template-url https:/ /s3.amazonaws.com/\
➥ awsinaction-code3/chapter13/multiaz.yaml \
➥ --parameters "ParameterKey=JenkinsAdminPassword,
➥ ParameterValue=$Password" \
➥ --capabilities CAPABILITY_IAM
```